Q.1.

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

```
class Demo {
    psvm() {
        int n=4;
        // uper pyramid
        for (int i=1; i<=n; i++) {
            for (int j=1; j<=n-i; j++) {
                sop(" ");
            }
            for (int k=1; k<=i; k++) {
                sop(k+" ");
            }
            sop(" ");
        }
        // lower pyramid
        for (int i=0; i<=n-1; i++) {
            for (int j=1; j<=i; j++) {
                sop(" ");
            }
            for (int k=1; k<=n-i-1; k++) {
                sop(k+" ");
            }
            sop(" ");
        }
    }
}
```

Dry Run -

i=1, i<=n
↳ j=1, j<=i

i=2 → j<=i
↳ 2

3 → j<=i
↳ 3

Q.2.

```
1                   1
1 2               2 1
1 2 3           3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3           3 2 1
1 2               2 1
1                   1
```

```
class Demo{
    psvm(){
        int n = 4
        int count = 3
        int sum;
        for (int i = 1; i<=n; i++){
            sum = i;
            for (int j = 1; j<=i; j++){
                sop( j+" ");
            }
            for (int j = 1; j<=2*(n-i); j++){
                sop(" " + " ");
            }
            for (int j = 1; j<=count; j++){
                sop(" ");
                count--;
            }
            for (int j = 1; j<=i; j++){
                sop(sum + " ");
                sum--;
```

```
            sop( );
        }
        // upside down

        for (int i=0; i<=n; i++) {
            sum = n-i;
            for (int j=1; j<=n-i; j++) {
                sop(j+" ");
            }
            for (int j= 1; j<=2*i; j++) {
                sop(" "+" ");
            }
            for (int j=1; j<=count ; j++) {
                sop(" ");
                count--;
            }
            for (int j=1; j<=n-i; j++) {
                sop(sum+" ");
                sum--;
            }
            sop( );
        }
    }
}
```

Q.3.

```
1     1
  2 2
   3
  4 4
5     5
```

```
int n = 5
For(int i=1; i<=n; i++){
  For(int j=1; j<=n; j++){
    if (i==j || i+j==n+1){
      sop(i);
    }
    else{
      sop(" ");
    }
  }
  sop();
```

Dry Run.

i=1; i<=n

↳ j=1; j<=n

↳ if(i=j ||i+...) {

↳ (" i ");

j<=n

↳ (else) {

↳ ("_ ")

A.S. 3.

Q.6

```
1
2 3
4 5 6
7 8 9 10
11 12 13
14 15
16
```

```
- Class Demo {
    psvm() {
        int n = 4;
        int count = 1;
    // upper pyramid
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n-i; j++) {
            sop(" ");
        }
        for (int k = 1; k <= i; k++) {
            sop(count + " ");
            count ++;
        }
        sop(" ");
    // lower pyramid
    for (int i = 1; i <= n-1; i++) {
        for (int j = 1; j <= i; j++) {
            sop(" ");
        }
        for (int k = 1; k <= n-i-1; k++) {
            sop(count + " ");
            count ++;
        }
        sop(" ");
    }
}
}
```

Dry Run -

n=4, count=1,

i=1; i<=n

↳ j=1; j<=i

↳ count

↳ 1

count++

2<=i

↳ num

↳ 2

num++

i=n-1, i>=1

↳ j=1; j<=1

↳ num

num++

Q.7.

```
   1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  0 1
   1
```

```
class Demo{
    psvml){
        int n=4;
    // uper pyramid
    for (int i=1; i<=n; i++){
        for (int j=1; j<=n-i; j++){
            sop(" ");
        }
        for (int k=1; k<=i; k++){
            if ((i+k) % 2==0){
                sop("1");
            }
            else{
                sop("0");
            }
        }
        sop(" ");
    }
    // lower pyramid
    for (int i=0; i<=n-1; i++){
        for (int j=1; j<=i; j++){
            sop(" ");
        }
        for (int k=1; k<=n-i-1; k++){
            if ((i+k) % 2==0){
                sop("0");
            }
            else{
                sop("1");
            }
        }
        sop(" ");
    }
}
}
```